Kamsi Nwabueze

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Education

University of Southern California Los Angeles, CA
B.S. in Computer Engineering and Computer Science *May 2028*

- Relevant Coursework: Computer Architecture, Embedded Systems, Digital Circuits, Parallel and Distributed Computation, Data Structures, Linear Algebra, Professional C++, Theory of Computation and Algorithms

Technical Skills

Languages: C, Embedded C, C++, Verilog, Python, Go,
Hardware & Embedded: FPGA Development ,UART, RS-232, Interrupts, Memory-Mapped I/O, Sensor Integration, CUDA
Tools: Vivado, Git, Linux, Bash, Docker

Work Experience

Vitel Wireless Limited Los Angeles, CA
Network Engineering Intern *June 2025 – August 2025*

- Designed and built Veye, a modular ONVIF-compatible home security platform integrating 4G IP cameras with cloud processing for large-scale remote monitoring
- Integrated the Shinobi open-source NVR using Node.js and Docker to manage and process 10,000+ concurrent video streams
- Built backend infrastructure enabling secure remote camera access across 50,000+ deployed VITEL 4G SIM cards

GistMe Inc Los Angeles, CA
Software Engineering Intern *June 2024 – August 2024*

- Developed a RESTful API using Flask for a web-based trivia platform supporting 10,000+ concurrent users
- Designed scalable NoSQL data storage, reducing average retrieval latency by 30%

Projects

Sable | *Verilog*

- Built an FPGA accelerator implementing the Power Iteration algorithm to compute dominant eigenvectors for Markov chain analysis
- Designed an FSM with BRAM-backed matrix and vector storage and epsilon-based convergence detection
- Implemented a 4-stage pipelined matrix–vector multiplication to accelerate iterative convergence
- Integrated a VGA visualization pipeline with clock-domain crossing FIFO and debounced hardware inputs

Atlas | *Verilog*

- Designed and implemented an 8-lane SIMD vector processing engine in Verilog to accelerate data-parallel workloads
- Built a vector register file and instruction decoder supporting parallel vector ALU operations across execution lanes
- Implemented modular RTL datapath components including ALU lanes, control FSM, and write-back logic
- Achieved 8× throughput over scalar execution via simultaneous multi-lane vector computation

Beluga | *Embedded C*

- Developed a microcontroller-based temperature monitoring system interfacing DS18B20 sensor, LCD display, RGB LEDs, servo motor, EEPROM, and UART peripherals
- Implemented interrupt-driven firmware coordinating sensor sampling, actuator control, and real-time display updates
- Designed an interrupt-based RS-232 protocol enabling configuration synchronization between devices
- Implemented persistent configuration storage using EEPROM with structured command parsing over serial interface

Sentinel | *Python, Raspberry Pi, GrovePi*

- Built an IoT monitoring system integrating ultrasonic distance sensors, rotary input, and RGB LCD display via GrovePi
- Implemented sensor polling and threshold detection to determine object proximity using ultrasonic ranging
- Mapped analog rotary input to configurable detection thresholds across a 10-bit sensor range
- Developed a real-time dashboard on an LCD interface displaying threshold values and object presence alerts

Orion | *Python, Raspberry Pi, Arduino Pro Mini*

- Implemented a multithreaded Chip-8 interpreter with opcode decoding, CPU execution cycle emulation, register state, and stack control flow
- Designed a deterministic execution loop with delay and sound timers synchronized to interpreter cycles
- Integrated Raspberry Pi GPIO input and interfaced an Arduino Pro Mini over UART for joystick ADC conversion
- Built a framebuffer renderer using bitwise sprite rasterization with SDL/OpenGL display output

Peregrine | *Go*

- Implemented an iterative DNS resolver using UDP sockets and raw DNS wire-format parsing (RFC 1035)
- Built a thread-safe TTL-based caching layer to reduce upstream query load
- Implemented concurrent request handling using goroutines and timeout-based retries

Leadership

Advanced Robotics Combat Los Angeles, CA
Project Lead *September 2025 – December 2025*

- Led a team of five engineers in the design and construction of a combat robot competing against 50+ teams
- Oversaw subsystem integration, CAD design, electrical architecture, and system testing while managing budgeting, engineering workflow, and competition strategy